Text-Based Prediction of Book Review Popularity

Goodreads Poetry | Extended paper-inspired experiment

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Reference: Bridget Daly, Stanford STATS229, Spring 2021

Prediction target

Classify whether a review receives more than 2% of its book's recorded review engagement.

Engagement = votes + comments

Share = review engagement / total engagement for the book
Popular when share > 0.02; unpopular otherwise

Example: 10 / 400 = 0.025 → popular

The label measures relative engagement; it does not measure positive sentiment or future likes.

Dataset and filtering

Step	Reviews retained
Raw reviews	154,555
Valid engagement counts	154,543
Positive engagement	59,193
Metadata review count ≥10	42,081
Book engagement ≥60	20,230
English confidence ≥0.90	9,663

9,663 reviews
462 books
6,336 users

2,604 popular
26.95% of final sample

Two compressed Poetry files joined on book_id. English filtering uses official fastText lid.176.bin.

Four feature sets

Set	Inputs	Features
A	User activity, review age, rating, rating difference	4
B	A + length, POS fractions, quotes, sentiment	12
C	B + Bag of Words (BoW)	2,012
D	B + TF-IDF	2,012

C/D: up to 2,000 vocabulary terms fitted on training reviews only

Activity and age are retrospective snapshot inputs. Direct votes, comments and share are not predictors.

Experiment design

Split	Reviews	Books
Train	7,374	322
Validation	1,184	70
Test	1,105	70

LR, XGBoost and a small MLP

A/B: full + balanced training
C/D: balanced training

18 experiments; 36 candidates

Balanced training: 3,766 reviews
Select parameters and final model by validation ROC-AUC

Seed 229. Numeric scaling and vocabularies fit training rows only; test data is used for reporting.

Held-out performance

Model	Accuracy	Recall	Specificity	AUC
Majority baseline	0.686	0.000	1.000	0.500
XGBoost B · full	0.753	0.398	0.916	0.768
XGBoost B · balanced	0.678	0.735	0.652	0.766
LR B · balanced	0.668	0.640	0.681	0.724
XGBoost D · balanced	0.684	0.726	0.665	0.772

Selected: XGBoost B, full training
Balancing raises recall and reduces accuracy at threshold 0.5

The model with the highest observed test AUC was not selected using that test score.

Logistic-regression feature rankings

C: Bag of Words

Feature	Coefficient
user_reviews	+0.806
num_words	+0.623
"kaur"	-0.573
"pg"	-0.511
"lovelace"	-0.505

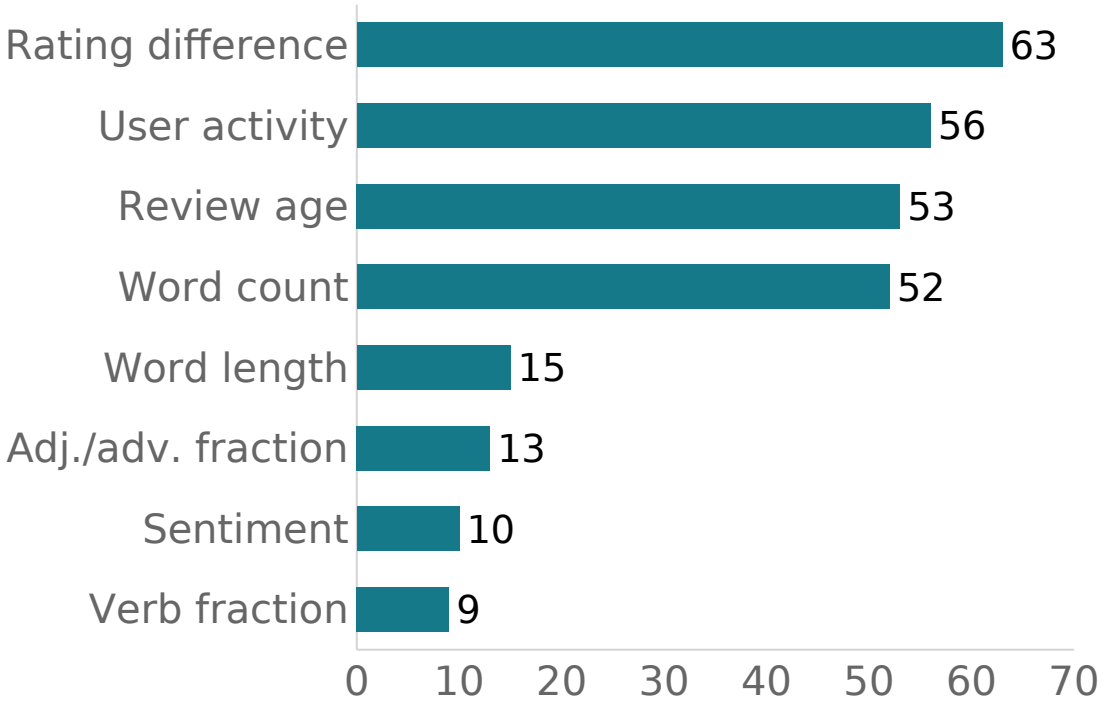
D: TF-IDF

Feature	Coefficient
user_reviews	+0.622
"kaur"	-0.486
"part"	-0.480
num_words	+0.408
"poet"	+0.393

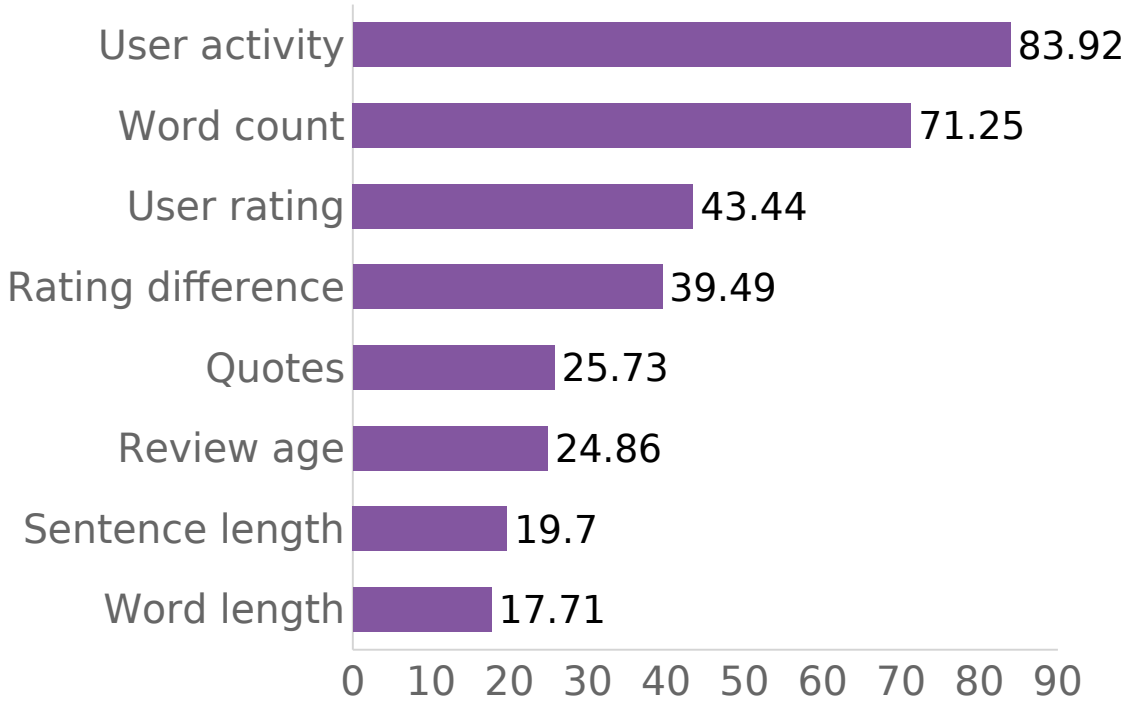
Top five by absolute magnitude. Signs show associations in this fitted representation; C/D units differ.

Subset B feature importance

XGBoost balanced: split counts



NN: mean |SHAP| × 1,000



Top eight inputs. NN values rescaled ×1,000 for display. Different importance units; no causal interpretation.

Scores and actual engagement share

Model	Pearson r	Spearman rho
LR B · balanced	0.366	0.413
XGBoost D · balanced	0.407	0.507

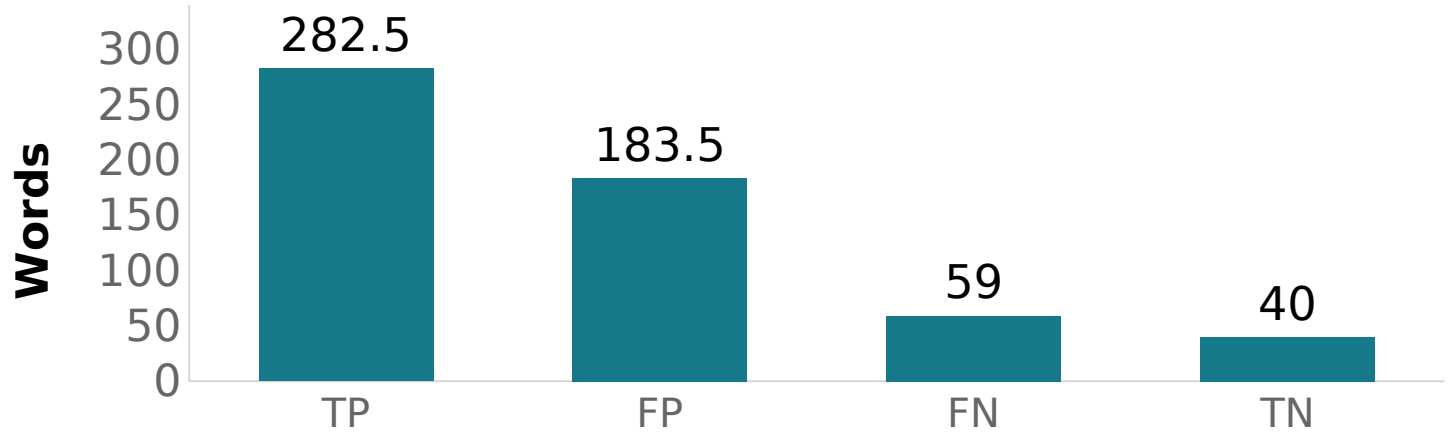
A score of 0.7 does not mean 70% of a book's engagement
These are binary classifiers, not engagement-share regressors

Scores correlate with share, but balancing changes prevalence and these scores are not calibrated.

Errors in XGBoost and balanced LR

Selected model	TP	FP	FN	TN
XGBoost B · full	138	64	209	694

LR B, balanced: median review length by error group



TP: popular found
FP: false alarm
FN: popular missed
TN: unpopular rejected

Length differences describe this split; they do not prove that longer reviews cause popularity.

Findings from the Poetry study

Validation chose XGBoost B with full training
On test books: AUC 0.768, recall 0.398, specificity 0.916
Balancing lifts recall to 0.735 and lowers accuracy to 0.678

In set B, tree splits rank rating difference first
In set B, NN SHAP ranks reviewer activity first

The inputs describe a collected Poetry snapshot
These associations do not establish future popularity or causation

Results use held-out books. Feature rankings describe fitted models, not causes.

Comparison with the full paper

Component	Paper	Poetry adaptation
Filtered reviews	About 1.98 million	9,663
Holdout	Random review splits	Entire books held out
Word representation	10,000-word BoW; larger TF-IDF	Up to 2,000 words each
A/B logistic regression	Unscaled, unregularized	Standardized, L2 penalty
XGBoost / neural net	Larger tree grid / Keras	100 trees / small sklearn MLP
Neural explanations	Large Deep SHAP sample	Smaller sampled Kernel SHAP

Major analysis categories are covered. Dataset, splitting, model settings and explanation budgets differ.

Appendix: all experiments (1/2)

Experiment	Val AUC	Test AUC	Accuracy	Recall	Specificity
LR A Full	0.646	0.713	0.715	0.133	0.982
XGB A Full	0.704	0.741	0.737	0.282	0.945
NN A Full	0.698	0.732	0.738	0.311	0.933
LR A Bal.	0.625	0.709	0.670	0.542	0.728
XGB A Bal.	0.687	0.730	0.662	0.692	0.648
NN A Bal.	0.684	0.728	0.690	0.622	0.720
LR B Full	0.672	0.736	0.727	0.231	0.954
XGB B Full	0.729	0.768	0.753	0.398	0.916
NN B Full	0.711	0.764	0.754	0.360	0.934

LR: logistic regression. XGB: XGBoost. NN: neural network. Bal.: balanced training.

Appendix: all experiments (2/2)

Experiment	Val AUC	Test AUC	Accuracy	Recall	Specificity
LR B Bal.	0.656	0.724	0.668	0.640	0.681
XGB B Bal.	0.718	0.766	0.678	0.735	0.652
NN B Bal.	0.692	0.746	0.696	0.723	0.683
LR C Bal.	0.634	0.701	0.656	0.637	0.665
XGB C Bal.	0.720	0.768	0.686	0.746	0.658
NN C Bal.	0.626	0.685	0.633	0.657	0.621
LR D Bal.	0.653	0.729	0.669	0.669	0.669
XGB D Bal.	0.722	0.772	0.684	0.726	0.665
NN D Bal.	0.646	0.711	0.629	0.692	0.600

LR: logistic regression. XGB: XGBoost. NN: neural network. Bal.: balanced training.